

TikTok AI Generator

Technical Specification

Automated TikTok Video Generation Pipeline

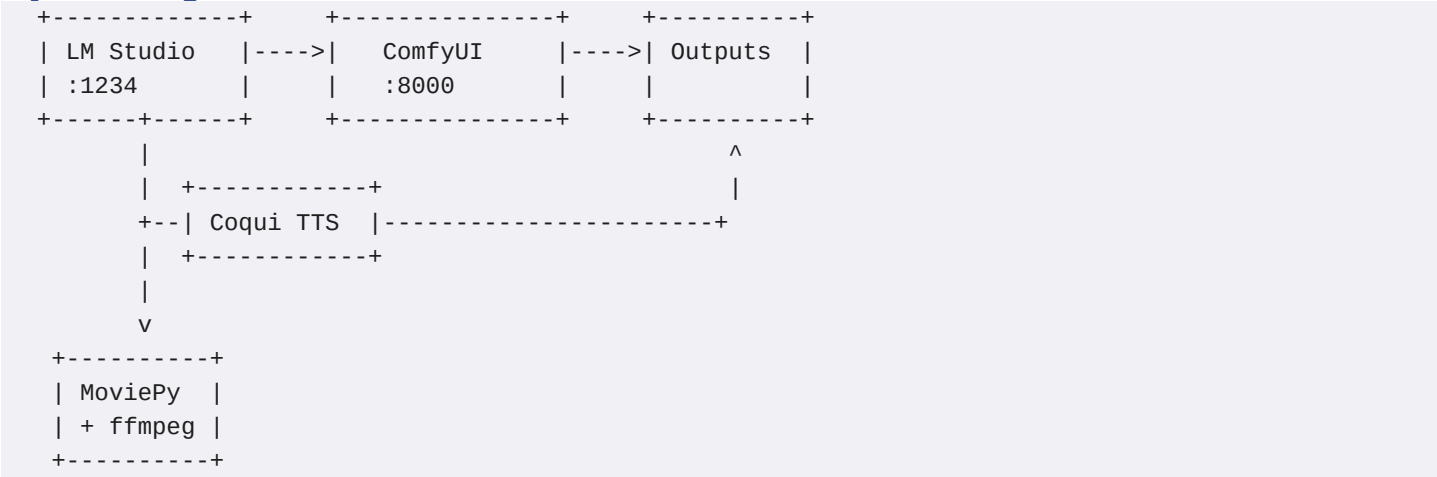
Prepared: May 2026

Stack: Python, LM Studio, ComfyUI, Coqui TTS, MoviePy

1. Architecture Overview

The system orchestrates multiple local AI services in a sequential pipeline to produce TikTok-format videos. Each stage feeds its output into the next, with no human intervention required between script generation and final video export.

System Diagram



2. Pipeline Stages

1. Script Generation

Function: `generate_script()`

Sends a system prompt (expert historian/TikTok scriptwriter) and user prompt to LM Studio via the OpenAI-compatible API. The model returns a JSON object with a 'script' key containing a <80 word historical curiosity in Spanish.

2. Image Prompt Generation

Function: `generate_image_prompt_from_script()`

Takes the script and sends a new prompt to LM Studio requesting a concise English image generation prompt (<30 words) optimized for photorealistic Stable Diffusion models.

3. Image Generation

Function: `generate_image_with_comfy()`

Loads a ComfyUI workflow JSON, injects the prompt into the CLIPTextEncode node (ID 6), and POSTs it to ComfyUI's /prompt endpoint. Awaits 3 output PNG images.

4. Text-to-Speech

Function: `text_to_speech()`

Uses Coqui TTS with the 'tts_models/es/css10/vits' model to synthesize Spanish speech from the script. Saves as 16-bit WAV.

5. Video Assembly

Function: `make_video_with_subs()`

Loads images as ImageClips with proportional duration based on audio length. Applies crossfade transitions (0.5s). Generates subtitle clips (Arial-Bold 60pt, white with black stroke, bottom-center). Composites everything with MoviePy.

6. Audio/Video Merge

Function: `merge_video_audio()`

Runs ffmpeg subprocess to remix the MoviePy output (which may have degraded audio) with the original WAV, encoding to AAC with libx264 video.

3. API Endpoints

The pipeline communicates with two local services via HTTP:

Service	URL	Method	Purpose
LM Studio	:1234/v1	POST	Chat completions
ComfyUI	:8000/prompt	POST	Image gen trigger

4. Model Configuration

Component	Model ID	Hardware
LLM	openai/gpt-oss-20b *	CPU/GPU (LM Studio)
TTS	tts_models/es/css10/vits	GPU (CUDA)
Image Gen	Absolute Reality v18.1 **	GPU (ComfyUI)

* Configurable via LM Studio model selection

** Via ComfyUI workflow; model loaded externally in ComfyUI

5. LLM Parameters

Parameter	Script Gen	Image Prompt Gen
Model ID	openai/gpt-oss-20b	openai/gpt-oss-20b
Temperature	0.8	0.8
Max Tokens	540	560
System Prompt	Historian + TikTok writer	Image prompt expert

6. ComfyUI Integration

The ComfyUI workflow is stored as a JSON template at workflows/Image Generation (Default).json. The pipeline modifies node ID 6 (CLIPTextEncode) to inject the generated prompt. Output images are saved to ComfyUI's output directory.

Image Discovery

Images are filtered by a PREFIX constant (python_development) in the filename. After submitting the prompt, wait_for_new_images() polls the output directory for up to 5 seconds until 3 matching PNG files appear.

7. Video Output Specifications

Parameter	Value
Container	MP4
Video Codec	libx264
Audio Codec	AAC
FPS	24
Bitrate	2000 kbps
Encoding Preset	medium
Transition	Crossfade (0.5s)
Subtitle Font	Arial-Bold, 60pt
Subtitle Style	White / black stroke
Subtitle Position	Bottom-center
Duration	Determined by audio

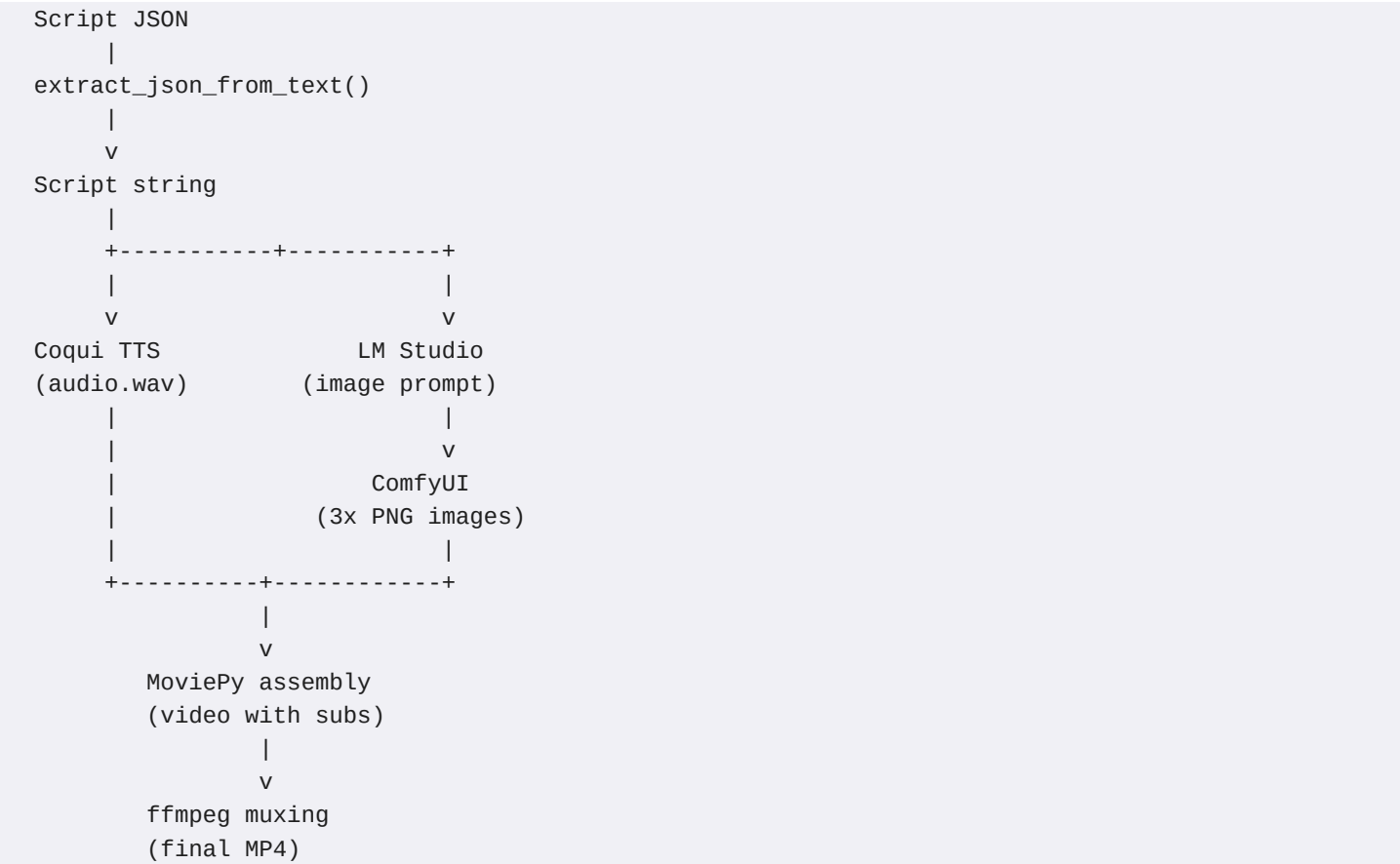
8. Dependencies

Library	Purpose
TTS (Coqui)	Text-to-speech inference
moviepy	Video editing and composition
openai	LM Studio API client
requests	ComfyUI HTTP client
ffmpeg	Audio/video muxing
ImageMagick	MoviePy text rendering
fpdf2	PDF generation (this doc)

9. Project Structure

```
.
+-- main.py                # Pipeline orchestrator (370 lines)
+-- workflows/
|   +-- Image Generation (Default).json # ComfyUI template
+-- outputs/                # Generated media (gitignored)
+-- .gitignore
+-- README.md
```

10. Data Flow



11. Key Configuration Points

Edit main.py to customize:

- COMFYUI_OUTPUT: Path to ComfyUI output directory for image discovery
- PREFIX: Filename prefix used to identify generated images
- Model names: LM Studio model ID and Coqui TTS model name
- crossfade_duration: Transition length between images (default 0.5s)
- Video codec, bitrate, and preset in write_videofile() calls